

Applying NLP for Topic Modelling in a Real-life Context

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1. Introduction

1.1 Business Context

PureGym Group is a leading low-cost gym operator offering flexible memberships and 24/7 access. In a highly competitive market, maintaining a strong customer experience is critical. Online reviews from platforms such as Google and Trustpilot provide valuable insight into customer satisfaction and improvement areas.

1.2 Object of the Analysis

This analysis applies NLP techniques to customer reviews to identify key themes, sentiment patterns, and drivers of negative experiences. Using methods such as topic modelling, emotion analysis, and LLMs, it focuses on uncovering critical pain points to support actionable recommendations for improving service quality and overall customer experience.

2. Data and Preprocessing

2.1 Data Sources and Cleaning

The analysis uses two datasets of customer reviews from Google and Trustpilot over a 12-month period, including review text, ratings, and location data. This enables both textual and location-based analysis, as well as cross-platform comparisons for overlapping locations. To ensure data quality, records with missing review text were removed and only valid, relevant entries were retained.

2.2 Text Preprocessing Techniques

With NLTK, the text was standardised by converting to lowercase, removing stopwords (including domain-specific terms such as “gym” and “puregym”), and eliminating numbers and non-informative characters. The cleaned text was then tokenised, preparing it for analysis such as frequency distribution, topic modelling, and emotion classification.

3. Exploratory Text Analysis

3.1 Word Frequency Analysis (All Reviews)

Word frequency analysis across Google and Trustpilot reviews highlights common terms such as “equipment”, “staff”, “great”, and “friendly”, indicating that feedback is primarily focused on facility quality, staff interactions, and overall gym experience.

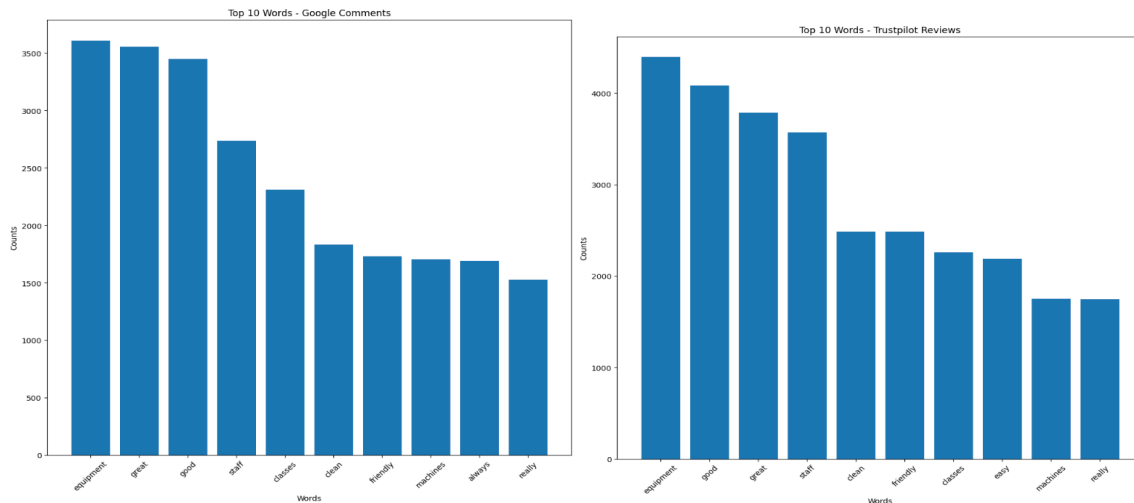


Figure 1. Word Frequency (Google Comments vs Trustpilot Reviews) – All Reviews

3.2 Word Cloud Visualisation (All Reviews)

Word clouds reinforce these findings, emphasising key themes such as equipment availability, cleanliness, and crowd levels. While positive terms appear frequently, the prominence of operational issues suggests they remain consistent areas of concern.



Figure 2. Word Cloud (Google Comments vs Trustpilot Reviews) – All Reviews

3.3 Analysis of Negative Reviews

The datasets were filtered to include only negative reviews (ratings below 3), enabling a more focused analysis of the key drivers of customer dissatisfaction.

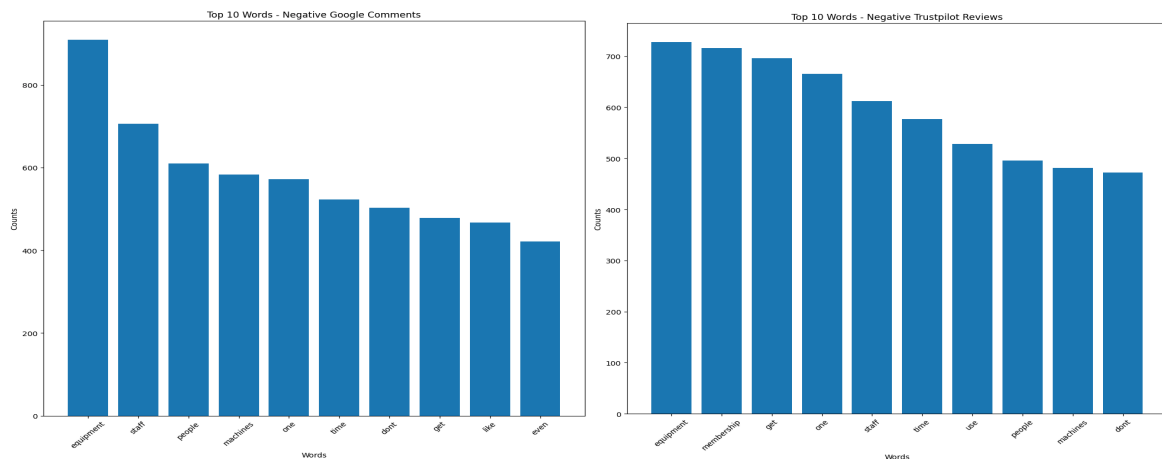


Figure 3. Word Frequency (Google Comments vs Trustpilot Reviews) – Negative Reviews



Figure 4. Word Cloud (Google Comments vs Trustpilot Reviews) – Negative Reviews

Frequency analysis and word clouds on negative reviews highlight terms such as “equipment”, “machine”, “shower”, “staff”, and “issue”, pointing to operational inefficiencies and service quality concerns. Compared to the full dataset, this shift suggests that negative experiences are driven mainly by tangible issues such as equipment maintenance, cleanliness, and capacity management.

4. Topic Modelling with BERTopic

4.1 Initial Topic Modelling

BERTopic (nr_topics = 15) was applied to negative reviews from locations common to both datasets to ensure consistent cross-platform comparison. The model identified distinct topics representing key drivers of negative sentiment.

Topic	Count	Name	Representation	Representative_Docs
1	0	1450	0_equipment_staff_machines_people	[equipment, staff, machines, people, air, park... [Recently got back to the gym after coming hom...
2	1	312	1_showers_shower_water_cold	[showers, shower, water, cold, hot, changing, ... [Hot water in the showers, Cold showers, Showe...
3	2	288	2_pin_pass_day_membership	[pin, pass, day, membership, code, access, did... [QR Codes in lobby don't work. Bought a day pa...
4	3	84	3_toilets_toilet_dirty_smell	[toilets, toilet, dirty, smell, changing, staf... [Very dirty! Equipment full of sweat, toilets ...
5	4	74	4_equipment_broken_machines_poor	[equipment, broken, machines, poor, missing, m... [not much equipment, Not enough equipment, Equ...
6	5	66	5_rude_staff_manager_unprofessional	[rude, staff, manager, unprofessional, managem... [Rude management :(, Very rude staff, especial...
7	6	46	6_wifi_signal_internet_months	[wifi, signal, internet, months, connection, t... [The wifi in the Exchange Quays Puregym is ter...
8	7	45	7_busy_kids_phones_way	[busy, kids, phones, way, machines, wait, time... [Busy, Too busy, Always busy]
9	8	29	8_crowded_overcrowded_round_cap	[crowded, overcrowded, round, cap, way, limit,... [Too much crowded, Too crowded, To crowded]
10	9	29	9_nah_didnt_past_know	[nah, didnt, past, know, walked, dont, dah, no... [Never been there in the 78 years of my life, ...
11	10	18	10_rubbish_good_hjollokkk_nis	[rubbish, good, hjollokkk, nis, terrible, wher... [Absolute rubbish, Absolutely rubbish, It's ju...
12	11	17	11_genocide_israel_supports_membership	[genocide, israel, supports, membership, ceo, ... [Too many students from two local colleges go ...
13	12	12	12_bench_bar_bars_benches	[bench, bar, bars, benches, safety, squat, fla... [Overcrowded and messy. Weights, dumbbells, an...
14	13	11	13_induction_booked_larger_person	[induction, booked, larger, person, min, showe... [This review is for my son. He went today for ...

Table 1. BERTopic on negative reviews with common locations – First run

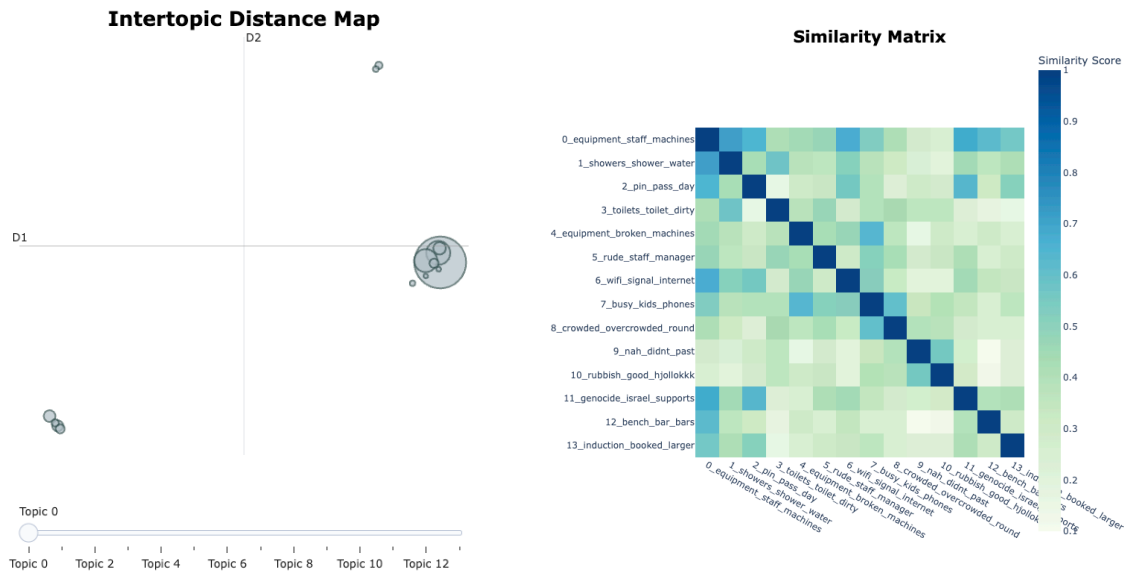


Figure 7. BERTopic Intertopic Distance Map and Similarity Matrix – First run

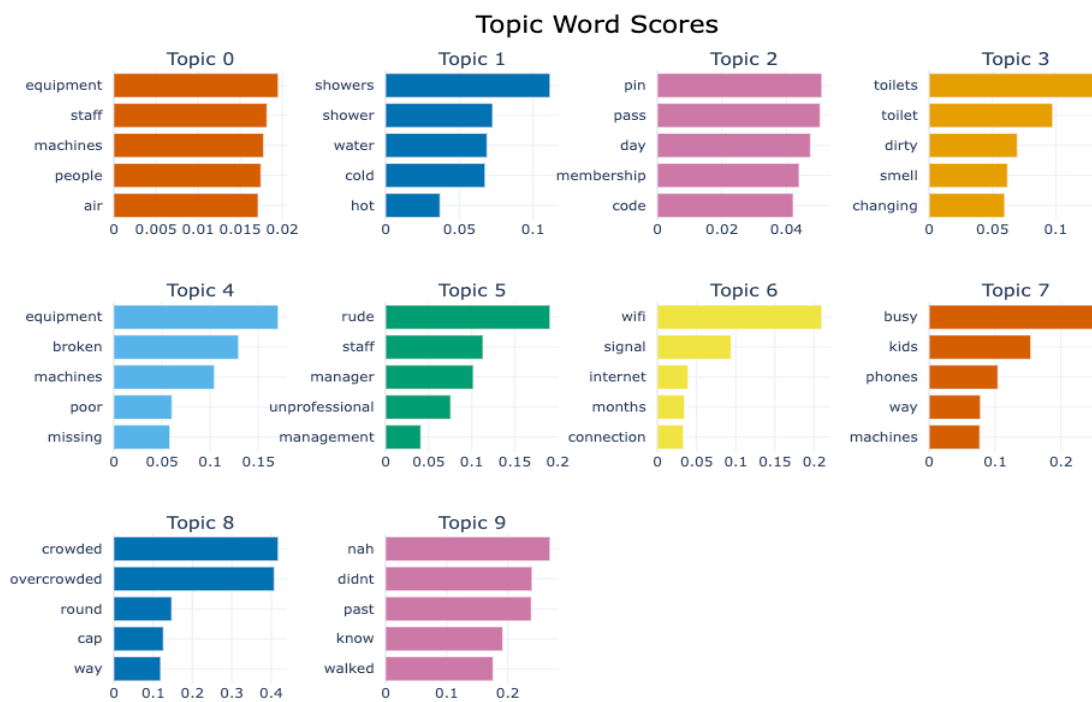


Figure 8. BERTopic Topic Word Series (Top 10) – First run

The results show well-separated topics, indicating feedback clusters around specific issues. Key themes include equipment availability, staff interactions, cleanliness, and access or membership processes, with strong consistency highlighting operational and service-related drivers of dissatisfaction. Further interpretation:

Topic	Title	Top words	Theme
0	Equipment & General Gym Experience	Equipment, staff, machines, people, air	Broad complaints about equipment availability/quality and overall gym

			experience, including interactions with staff and environment
1	Shower Facilities & Temperature	Showers, shower, water, cold, hot	Issues related to showers, particularly water temperature inconsistencies and usability
2	Access & Membership Systems	Pin, pass, day, membership, code	Problems with entry systems and membership access, including PIN codes and login issues
3	Toilets & Cleanliness	Toilets, dirty, smell, changing	Complaints about hygiene and cleanliness, especially in toilet and changing areas
4	Broken or Poor Equipment	Equipment, broken, machines, poor, missing	Focus on faulty, unavailable, or poorly maintained equipment
5	Staff Behaviour & Professionalism	Rude, staff, manager, unprofessional	Negative feedback regarding staff attitudes and professionalism, including poor customer service
6	WiFi & Connectivity Issues	WiFi, signal, internet, connection	Issues with internet connectivity, including weak signal and unreliable WiFi
7	Gym Environment & Disruptions	Busy, kids, phones, way, machines	Complaints about the gym environment, including distractions, inappropriate behaviour, and overcrowding
8	Overcrowding & Capacity	Crowded, overcrowded, cap, way	Issues related to overcrowding and limited space, leading to poor user experience
9	General / Informal Complaints	Nah, didnt, know, walked	Unstructured or general dissatisfaction, often expressed informally without a specific issue

Table 2. BERTopic Top 10 Topic Interpretation – First run

4.2 Topic Modelling for Top Locations

The first run captures a broad range of issues across the full dataset, including both operational concerns (equipment, staff, crowding) and platform-specific themes such as membership on Trustpilot. In contrast, the second run, focused on the top 30 locations, shows a more concentrated pattern, with less emphasis on administrative issues and a stronger focus on core operational problems.

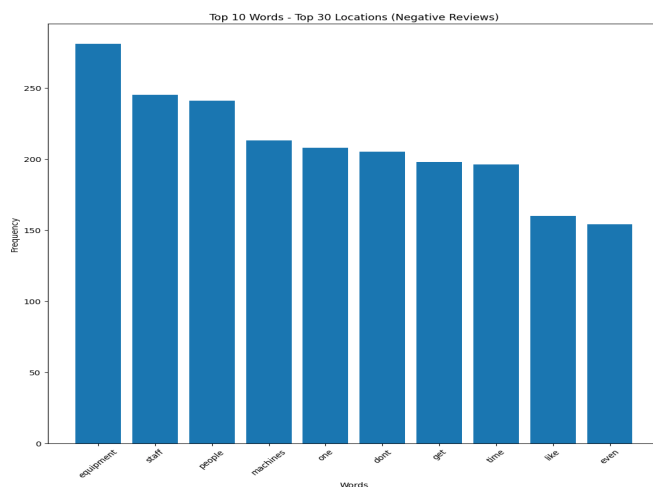


Figure 9. Word Frequency – Combined Negative Top Locations

Word frequency and word cloud results remain consistent across runs, but the second run highlights a narrower set of recurring issues, primarily equipment, staff, and overall gym experience, suggesting that dissatisfaction in high-traffic locations is driven by fewer, high-impact problems.



Figure 10. Word Cloud – Combined Negative Top 30 Locations

This shift is further reflected in BERTopic results. While the first run identifies a diverse set of topics (e.g. hygiene, facilities, access, connectivity), the second run produces fewer dominant themes, mainly centred on equipment, staff, and parking. Overall, this indicates reduced topic diversity and shows that issues in busier locations are more concentrated and operational in nature.

	Topic	Count	Name	Representation	Representative_Docs
1	1	918	1_equipment_staff_people_machines	[equipment, staff, people, machines, dont, jus...	[Pros: - It's a new gym, so everything is new,...
0	0	41	0_parking_car_free_85	[parking, car, free, 85, park, fines, reg, fin...	[Get free parking there in the car park it wou...

Table 3. BERTopic on combined negative reviews top 30 common locations – Second run

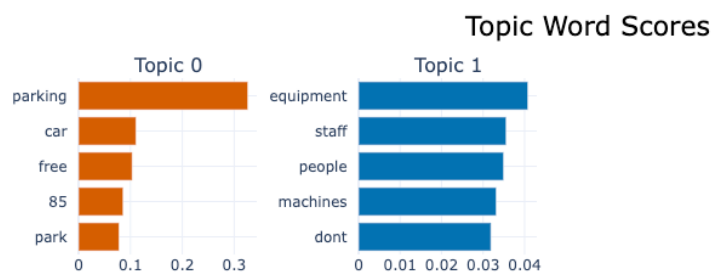


Figure 11. BERTopic Topic Word Series (Top 10) – Second run

5. Emotion Analysis and Further Topic Modelling

5.1 Emotion Classification using BERT

Emotion analysis was performed using a pre-trained BERT-based model (bhadresh-savani/bert-base-uncased-emotion) to classify the dominant emotion in each review (e.g. anger, fear, sadness, joy) across both datasets, providing insight into how customers felt about their experiences.

5.2 Emotion Distribution in Negative Reviews

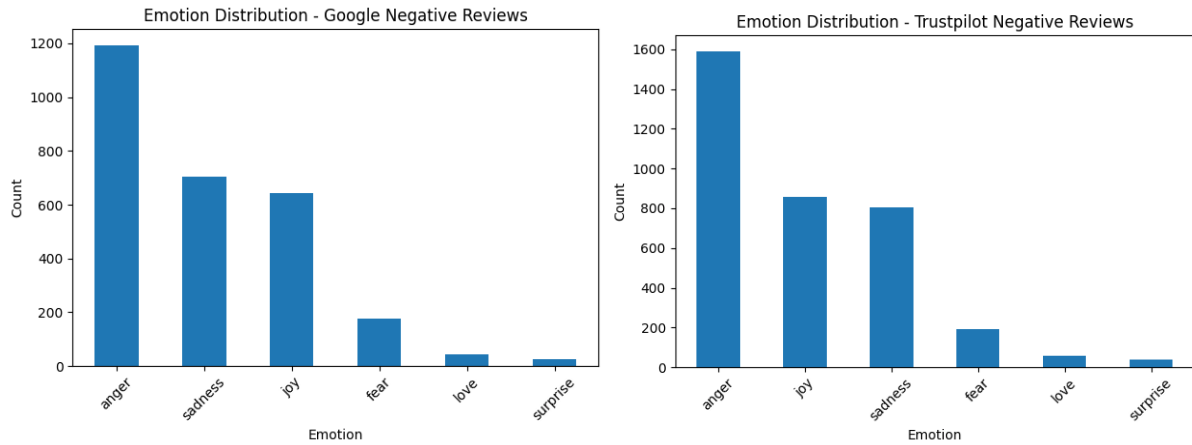


Figure 12. Emotion Distribution for Google & Trustpilot Negative Reviews

Emotion analysis shows anger as the dominant response, followed by sadness, indicating strong dissatisfaction. This is consistent across platforms, while “joy” likely reflects mixed sentiment, sarcasm, or model limitations.

5.3 Topic Modelling on Anger Reviews with BERTopic

Topic modelling using BERTopic was applied to the subset of angry reviews to identify the most critical issues.

Topic	Count	Name	Representation	Representative Docs
1	0	0_equipment_people_staff_just	[equipment, people, staff, just, machines, wei...	[Recently got back to the gym after coming hom...
2	1	1_membership_parking_class_time	[membership, parking, class, time, classes, da...	[I paid for 7 day pass and didn't receive a Pl...
3	2	2_fee_membership_joining_payment	[fee, membership, joining, payment, charged, c...	[There should be no joining fee, A company tha...
4	3	3_og_er_der_ikke	[og, er, der, ikke, det, und, til, man, jeg, die]	[Indeklimaet er så dårligt at man næsten ikke ...
5	4	4_showers_cold_water_shower	[showers, cold, water, shower, hot, temperatur...	[Showers COLD, Cold showers, Showers are too c...
6	5	5_equipment_busy_crowded_machines	[equipment, busy, crowded, machines, sb, peopl...	[Too crowded, not enough equipment, Too busy c...
7	6	6_rude_staff_manager_unprofessional	[rude, staff, manager, unprofessional, managem...	[Faulty equipment is not fixed for long period...
8	7	7_air_wifi_conditioning_ac	[air, wifi, conditioning, ac, hot, summer, goo...	[They don't turn the air conditioning on, Air-...
9	8	8_toilets_toilet_disgusting_changing	[toilets, toilet, disgusting, changing, room, ...	[Stop the cleans from sleeping in male toilets...
10	9	9_nah_past_didnt_walked	[nah, past, didnt, walked, havent, recommend, ...	[walked past, Never been there in the 78 years...
11	10	10_pin_sent_didnt_send	[pin, sent, didnt, send, did, took, app, ages,...	[Didn't give me a pin, I don't get the pin, Di...
12	11	11_code_work_sent_works	[code, work, sent, works, didnt, couldnt, jt, ...	[Can't get code, Code didn't work, the code wo...
13	12	12_phone_customer_service_speak	[phone, customer, service, speak, answer, numb...	[Doesn't have any customer service via phone a...
14	13	13_induction_booked_showed_waited	[induction, booked, showed, waited, class, tur...	[The online joining process was OK but on the ...

Table 4. BERTopic on combined anger reviews – Third run

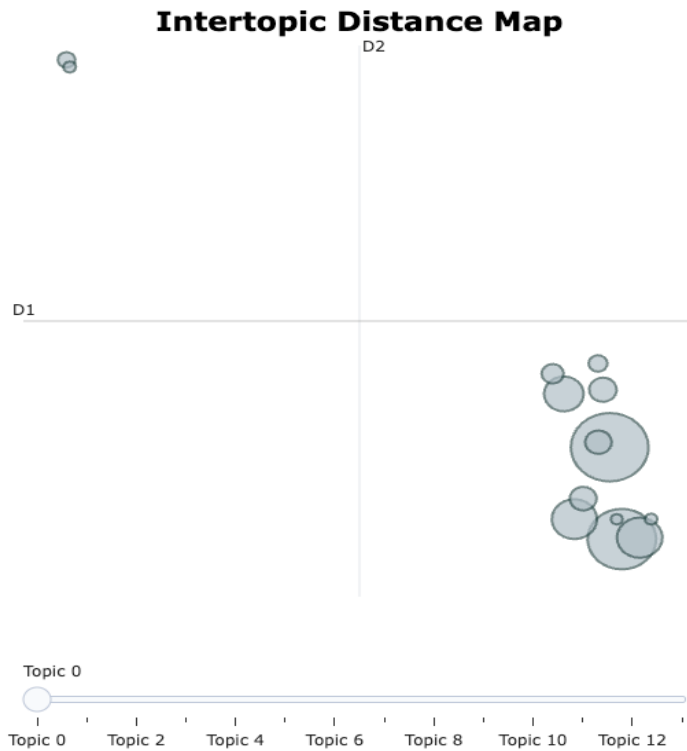


Figure 13. BERTopic Intertopic Distance Map – Third run

The third run shows a more concentrated clustering pattern, with a few dominant clusters and several smaller ones, indicating that anger-driven feedback is less diverse and centred on core issues such as equipment, staff behaviour, access or membership, and facilities.

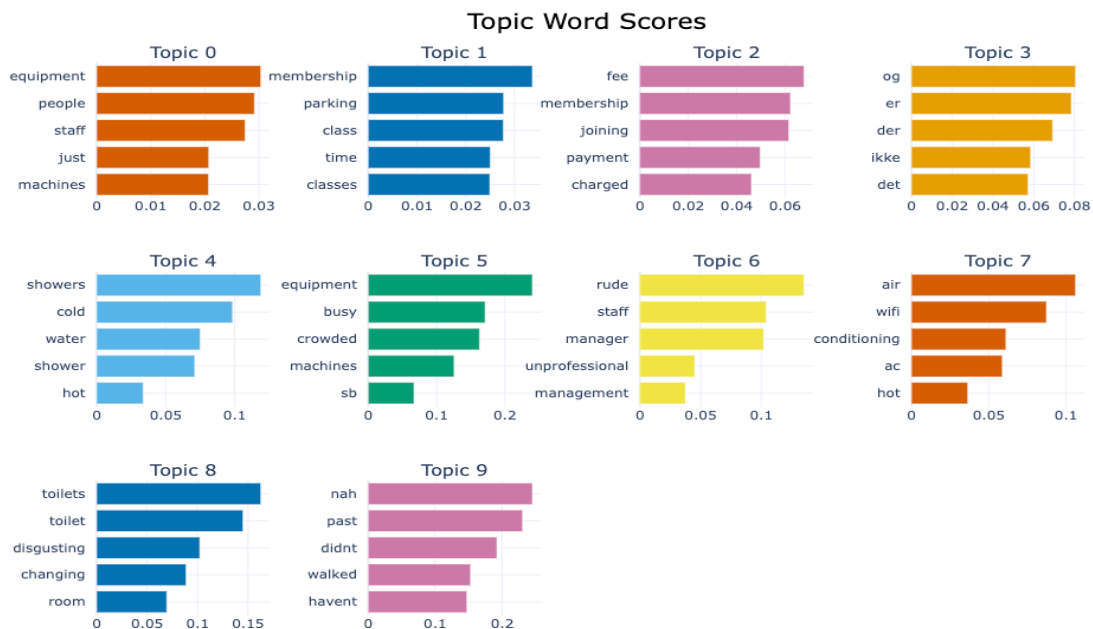


Figure 14. BERTopic Topic Word Series (Top 10) – Third run

The third run reintroduces multiple topics but with less clarity due to more informal and emotional language. It highlights key drivers of anger (e.g. membership issues, equipment failures, poor staff interactions, hygiene, and overcrowding), showing that strong negative emotions stem from specific high-impact problems.

6. Large Language Model (LLM) Analysis

6.1 Topic Extraction using Falcon-7B-Instruct

Falcon-7B-Instruct was used to extract the top three topics from each negative review, enabling more context-aware and human-readable insights compared to traditional keyword-based topic modelling.

6.2 BERTopic on Extracted Topics

The topics generated by the LLM were aggregated into a single dataset and used as input for BERTopic. This second-stage modelling aimed to identify overarching themes across the LLM-extracted topics.

Topic	Count	Name	Representation	Representative Docs
3	2	44 2_facilities_cleaning_hygiene_cleanliness	[facilities, cleaning, hygiene, cleanliness, e...	[Poor Facilities, Poor facilities, Bad hygiene...
1	0	37 0_customer_service_poor_nothing	[customer, service, poor, nothing, works, lack...	[Poor customer service, Poor customer service,...
7	6	37 6_equipment_machines_outdated_of	[equipment, machines, outdated, of, limited, l...	[Equipment, Equipment, Equipment]
8	7	35 7_gym_the_and_like	[gym, the, and, like, incline, for, if, needs,...	[Poor gym equipment, Gym, I have been using th...
2	1	24 1_staff_and_unhelpful_management	[staff, and, unhelpful, management, unfriendly...	[Staff, Staff, Staff]
6	5	21 5_service_customer_delivery_incomplete	[service, customer, delivery, incomplete, all,...	[Customer Service, Customer Service, Customer ...
4	3	19 3_quality_product_food_expensive	[quality, product, food, expensive, price, the...	[Product quality, Poor product quality, Qualit...
5	4	12 4_what_type_membership_on	[what, type, membership, on, unfair, cancellat...	[Unclear on what type of day trials., Unclear ...

Table 5. BERTopic on combined negative reviews using Falcom LLM – Fourth run



Figure 15. BERTopic Topic Word Series (Top 10) – Fourth run

The fourth BERTopic run, enhanced with Falcon LLM guidance, improves interpretability by producing clearer, more meaningful themes (e.g. customer service, cleanliness, pricing, service

delivery) instead of keyword-based clusters. It also consolidates similar issues and reduces noise, resulting in more structured, actionable insights aligned with key business problem areas.

6.3 Business Insights and Recommendations

The aggregated topics were then passed back into the Falcon-7B-Instruct model with a prompt requesting actionable recommendations. After some prompt engineering, below is the raw output of Falcon:

```
- Improve the customer service
- Improve the hygiene and cleanliness of the gym
- Upgrade the equipment
- Communicate with members
- Increase the range of gym products and services
- Provide more variety of workouts
- Offer more personalized workout plans
- Add new classes and activities
- Provide more flexibility in gym schedules
- Create more exciting gym events and promotions
- Offer unique gym rewards and incentives
- Increase accessibility and visibility of gym services
- Simplify the gym membership fees and
```

Image 1. Raw Falcon LLM Output for Business Recommendations

Which can be consolidated and refined to 5 main business recommendations:

Business Insights	Actions
Improve Customer Service and Communication	Enhance staff training and professionalism while improving communication around membership, billing, and policies. This addresses both service-related complaints and frustration caused by unclear or poor communication.
Upgrade Equipment and Maintain Facilities	Invest in regular maintenance and upgrades of gym equipment, while ensuring high standards of cleanliness across facilities such as toilets, showers, and changing areas. These are core drivers of dissatisfaction.
Enhance Overall Gym Experience and Environment	Reduce overcrowding, improve scheduling flexibility, and create a more comfortable workout environment (e.g. temperature, noise levels). This improves usability and member satisfaction.
Expand and Personalise Offerings	Introduce a wider variety of classes, workouts, and personalised fitness plans to better cater to different user needs and increase engagement.
Simplify Membership and Increase Perceived Value	Make pricing structures more transparent and introduce incentives, rewards, or promotions to improve value perception and member retention.

Table 6. Refined Business Recommendations – through Falcon output

7. Topic Modelling with Gensim

7.1 Latent Dirichlet Allocation (LDA)

LDA (Gensim) was applied to negative reviews, generating 10 topics based on word co-occurrence. It identified key themes such as equipment issues, cleanliness, overcrowding, and staff concerns, with some topics capturing membership issues and noise. Overall, LDA captured the main themes but produced broader, less interpretable results than BERTopic.

7.2 Visualisation

Visualisation using pyLDAvis showed moderate separation between topics, with some overlap indicating that issues are interconnected. For example, equipment problems often co-occur with overcrowding, while staff issues relate closely to overall service quality.

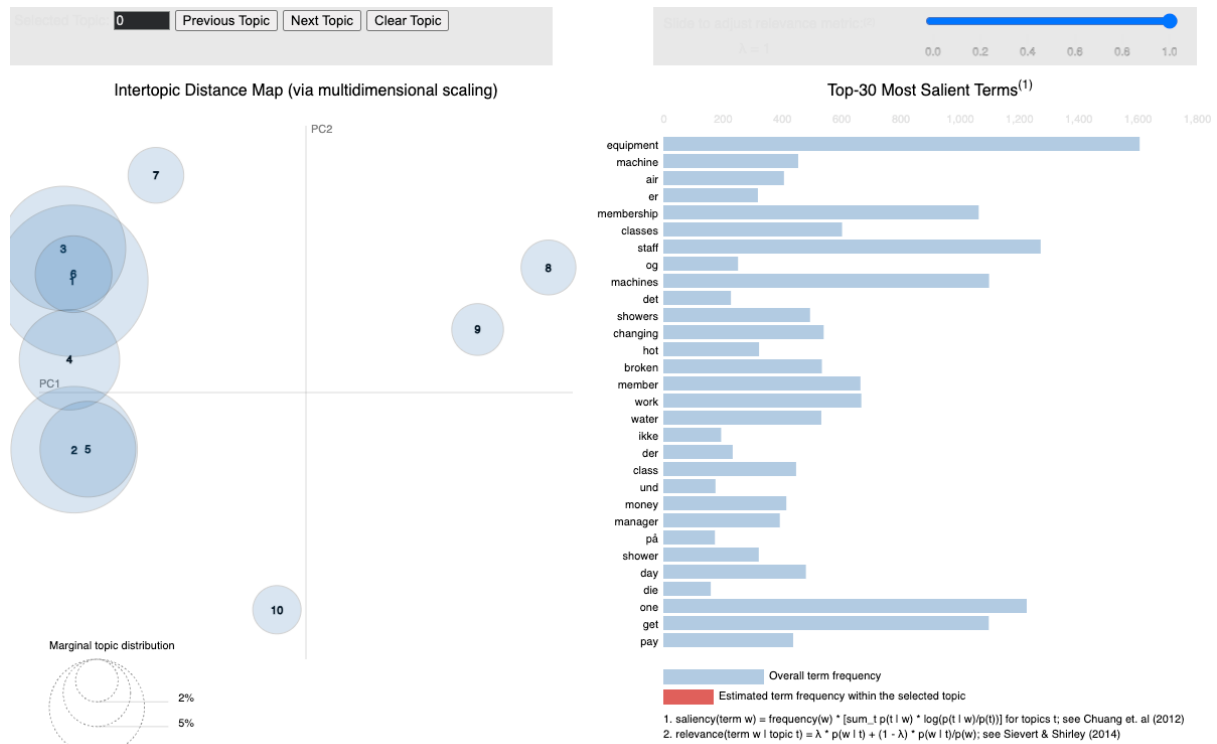


Figure 16. LDA Visualisation (pyLDAvis)

The most salient terms included equipment, machines, broken, staff, clean, membership, and showers, confirming that operational issues are the main drivers of dissatisfaction. Compared to BERTopic, LDA topics are less distinct and coherent, but the themes remain consistent across methods, including Falcon outputs, reinforcing the reliability of the findings.